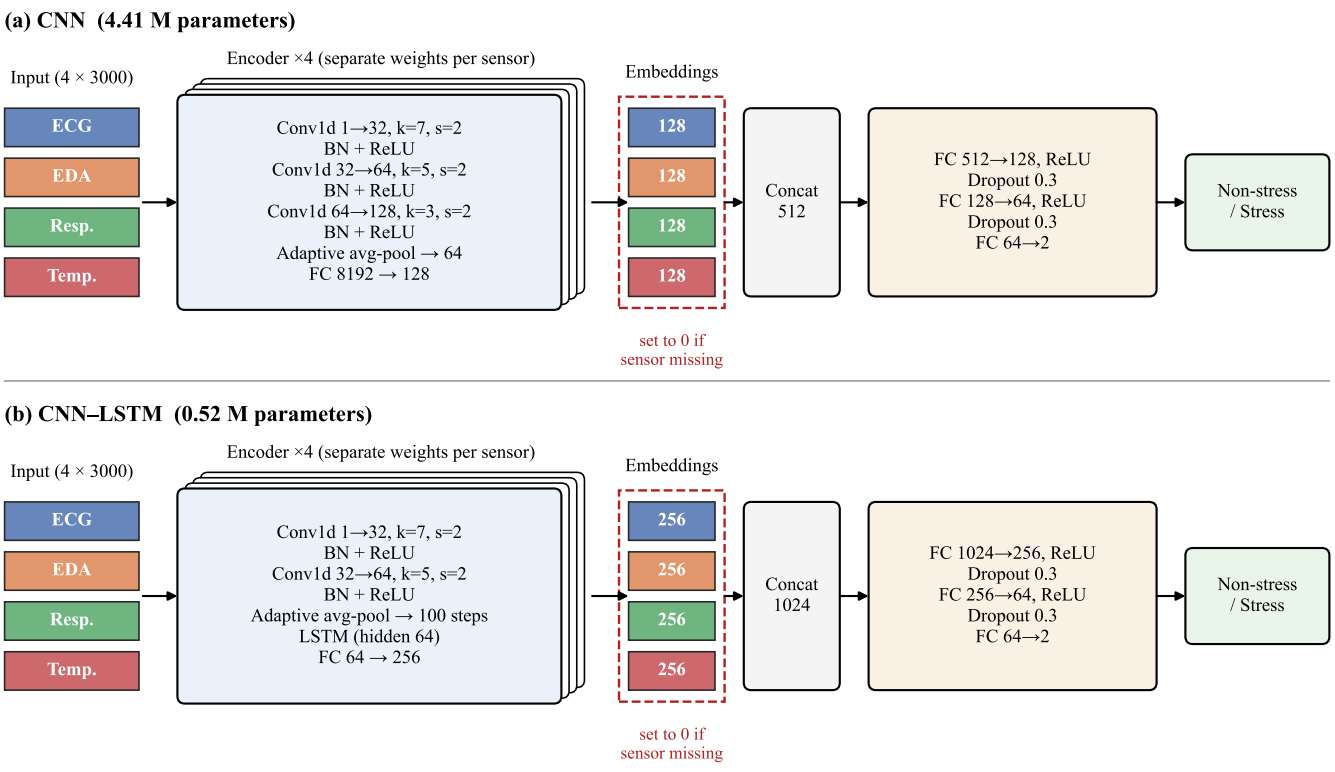
(a) CNN (4.41 M parameters)
Input (4 × 3000)
ECG
EDA
Resp.
Temp.
Encoder ×4 (separate weights per sensor)
Conv1d 1→32, k=7, s=2
BN + ReLU
Conv1d 32→64, k=5, s=2
BN + ReLU
Conv1d 64→128, k=3, s=2
BN + ReLU
Adaptive avg-pool → 64
FC 8192 → 128
Embeddings
128
128
128
128
set to 0 if sensor missing
Concat 512
FC 512→128, ReLU
Dropout 0.3
FC 128→64, ReLU
Dropout 0.3
FC 64→2
Non-stress / Stress
(b) CNN–LSTM (0.52 M parameters)
Input (4 × 3000)
ECG
EDA
Resp.
Temp.
Encoder ×4 (separate weights per sensor)
Conv1d 1→32, k=7, s=2
BN + ReLU
Conv1d 32→64, k=5, s=2
BN + ReLU
Adaptive avg-pool → 100 steps
LSTM (hidden 64)
FC 64 → 256
Embeddings
256
256
256
256
set to 0 if sensor missing
Concat 1024
FC 1024→256, ReLU
Dropout 0.3
FC 256→64, ReLU
Dropout 0.3
FC 64→2
Non-stress / Stress